

Summary

I used the ChatGPT 5.1 Thinking model under study mode to help go over pre-requisite content found in Homework 0. I told the model that they are a helper and the goal is to refresh the content found in homework 0 but now actually solve any of the problems. The first thing the model did was ask how comfortable I was with the pre-requisite content, which I appreciated since it tailored the difficulty of the questions from it. Afterwards, they kept on asking questions and explained why some of my answers were incorrect. I believe this is a good prompt to help students with the linear algebra content and basic ML concepts before diving into Homework 0.

Me: You are model helping me warm up for my homework assignment. This homework assignment contains prerequisite information I should know before starting the class but I need a refresher on these concepts. Do not provide homework answers or directly solve any assigned problems. Only ask and explain concepts. I have attached the homework assignment below. (attached homework 0)

ChatGPT:

Great — since you want a **warm-up tutor**, not homework answers, we'll spend our time **reviewing concepts, asking guiding questions, and checking your intuition**.

I'll use the homework PDF only to see **which topics** you need to know, not to solve any of its problems. (Referencing HW 0: vector calculus, SVD, least squares, ridge regression, ReLU behavior, etc.)

Before we dive in — quick check-in

I really liked how they checked how well the user understands the content before asking questions

To follow the Study Mode rules, I should first understand your background so I can scaffold effectively.

Can you tell me two things:

1. How comfortable are you *right now* with linear algebra topics like SVD, matrix calculus, and least squares?
(e.g., "very rusty", "mixed", "pretty confident")
2. How comfortable are you with neural network basics like ReLU, gradients, and SGD?

Once I know that, I'll build a structured warm-up path that reviews the prerequisite ideas **without touching any homework-specific solutions**.

Whenever you're ready, answer those two questions and we'll start with the foundational concept that you most need refreshed.

Me: I am very rusty with all these topics

ChatGPT:

Perfect — then we'll treat this like a **full refresher boot-up**, but in a guided, step-by-step way so that when you actually start HW0, nothing feels unfamiliar.

Let's begin with the absolute foundation that shows up *everywhere* in the assignment: **vector/matrix derivatives**.

(These appear repeatedly in Q2 of the homework.)

Instead of going over specific homework questions, they highlight concepts that will be covered and exactly where a student can find it in the homework

I'll start with a simple idea and ask you just **one small question at a time** so we can rebuild your intuition.

◆ Warm-Up Concept: What does a derivative *mean* for vectors?

When we take a derivative like

$$\left[\frac{\partial}{\partial x} (x^T c), \right]$$

we're not taking a derivative of a number with respect to a single variable — we're taking the derivative of a scalar with respect to an **entire vector**.

A helpful mental model:

- If the output is a **scalar**, and the input is an **n-dimensional vector**,
→ the derivative is a **row vector of length n**.

This matches the homework's convention:

$$\left[\frac{\partial f}{\partial x} = \text{bigg} \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right] \text{bigg} \right]$$

(also described in HW0 text on vector derivatives).

👉 Your first tiny warm-up question

Let's start very simple.

Suppose

[

$$f(x) = x^T c$$

]

with $(x, c) \in \mathbb{R}^n$.

Before doing *any* calculation:

Question 1 (conceptual):

What type of object should $\left(\frac{\partial f}{\partial x}\right)$ be?

- A scalar
- A column vector
- A row vector
- A matrix

Just pick which one you think it should be — don't worry about being correct; this is just to gauge where you are.

Me: column vector

ChatGPT:

Great — and that instinct is super common!

Let's refine it so you're aligned with the convention the homework uses.

♦ Quick correction + intuition

For a scalar function $f(x)$ where $(x \in \mathbb{R}^n)$, the homework explicitly adopts the convention:

$$\left[\frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right] \right]$$

— a **row vector**.

(This is stated in the HW text discussing vector derivatives.)

keeps the definitions consistent

Why?

Because a **row vector** times a **column vector** gives a **scalar**.

This keeps the shapes consistent.

So the answer under this convention is: **a row vector**.

No worries — this is one of the first rusty spots everyone has.

Next small step

Let's stay with the same function:

$$\begin{aligned} &[\\ f(\mathbf{x}) &= \mathbf{x}^T \mathbf{c} = \sum_{i=1}^n x_i c_i. \\ &] \end{aligned}$$

Question 2:

Without computing the full derivative, look at the summation form.

What is

$$\begin{aligned} &[\\ &\frac{\partial f}{\partial x_k} \\ &] \end{aligned}$$

for a single coordinate (x_k)?

builds up the difficulty and does a good job breaking things down

Just tell me what you think it is in terms of (c_k).

(Don't worry about the whole vector yet — just one entry.)

Me: c_k

ChatGPT:

Exactly right — nice!

$$\begin{aligned} &[\\ &\frac{\partial f}{\partial x_k} = c_k. \\ &] \end{aligned}$$

That's all you need to know to rebuild the whole derivative.

Since every coordinate derivative is (c_k), the full derivative is the row vector:

$$\left[\frac{\partial f}{\partial x} = [c_1, c_2, \dots, c_n] = c^T. \right]$$

This matches the expected structure in HW0 Q2(a).

You're already rebuilding the foundational intuition — good sign.

◆ Next Warm-Up Concept: Derivative of a quadratic norm

This shows up all over deep learning (including gradient descent, ridge regression, etc.).

Consider:

$$\left[f(x) = \|x\|^2 = x^T x = \sum_{i=1}^n x_i^2. \right]$$

We'll approach this step by step.

👉 Question 3

Looking at the summation:

$$\left[f(x) = \sum_{i=1}^n x_i^2, \right]$$

What is

$$\left[\frac{\partial f}{\partial x_k} \right]$$

Just answer for a **single coordinate** again, like before.

(We'll assemble the full vector afterward.)

Me: $2x_k$

ChatGPT:

Perfect — exactly right.

$$\left[\frac{\partial f}{\partial x_k} = 2x_k. \right]$$

Now let's assemble the full derivative.

Since each coordinate contributes $(2x_k)$, the full derivative under the homework's row-vector convention is:

$$\left[\frac{\partial f}{\partial x} = 2x^T. \right]$$

This completes the intuition for HW0 Q2(b).

You're doing great — the rusty pieces are coming back fast.

◆ Next Concept: Derivative of a matrix–vector product

This shows up constantly when computing gradients of neural nets and linear models.

Consider:

$$\left[f(x) = Ax, \right]$$

where

- $(A \in \mathbb{R}^{n \times n})$,
- $(x \in \mathbb{R}^n)$,
- and the output $(f(x))$ is a **vector**, not a scalar.

This is different from before because we're now taking the derivative of a **vector-valued function**.

👉 **Before we compute anything, here's your next guiding question:**

Question 4:

If $f(x) = Ax$ outputs an **n-dimensional vector**, and we take

[
 $\frac{\partial f}{\partial x}$,
]

what *shape* should the derivative have?

Pick one:

- A scalar
- A row vector
- A column vector
- An $(n \times n)$ matrix

(Think: a vector derivative of a vector—what should that produce?)

Me: matrix

ChatGPT:

Exactly — good instincts again.

When a function outputs a **vector**, its derivative with respect to a vector is a **matrix**.

More precisely, it's the **Jacobian**: each row is the gradient of one output component with respect to the input (x) .

For $f(x) = Ax$, this Jacobian will end up being... well, we'll discover that together.

◆ Next tiny step: build intuition through one coordinate

Let's write the function in coordinates:

$$[f_i(x) = \sum_{j=1}^n A_{ij} x_j.]$$

So the (i)-th output is a linear combination of the entries of (x).

Now consider what an entry of the Jacobian looks like:

$$[\frac{\partial f_i}{\partial x_k}.]$$

This is:

- “How does the (i)-th output change when I tweak the (k)-th input?”

Since $(f_i(x))$ is a sum over (j), only the term with $(j = k)$ survives.

👉 Question 5:

Given

$$[f_i(x) = \sum_{j=1}^n A_{ij} x_j,]$$

What is

$$[\frac{\partial f_i}{\partial x_k} ?]$$

Try to express it in terms of one entry of (A).

(Just like you did with (c_k) earlier.)

Me: A_{ik}

ChatGPT:

Exactly — you nailed it again.

$$\left[\frac{\partial f_i}{\partial x_k} = A_{ik} \right]$$

This means the Jacobian matrix has entries:

$$\left[\frac{\partial f}{\partial x} \right]_{ik} = A_{ik}.$$

So the entire derivative is simply:

$$\frac{\partial f}{\partial x} = A.$$

This matches HW0 Q2(c).

You're building momentum — great sign.

◆ Next Concept: Quadratic form ($x^T A x$)

This shows up everywhere in optimization, least squares, ridge regression, etc.

And HW0 Q2(d–e) asks about this explicitly.

Let's guide you into it step by step.

We look at:

$$f(x) = x^T A x.$$

This is a scalar (since $x^T A x$ is just a number).

Before touching any formulas, let's rewrite it in summation form to see the structure:

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j.$$

That makes it much easier to differentiate.

Question 6:

Focus on a single coordinate derivative again.

Given:

$$f(x) = \sum_{i,j} x_i A_{ij} x_j,$$

what terms in the sum include (x_k) ?

There are *two different types* of terms:

1. Terms where **(x_k) appears as the left multiplier:**
 $(x_k A_{kj} x_j)$
2. Terms where **(x_k) appears as the right multiplier:**
 $(x_i A_{ik} x_k)$

Your task:

List the two types of terms in words, something like:

- “Terms where $(i = k)$...”
- “Terms where $(j = k)$...”

Just describe them — no derivative yet.